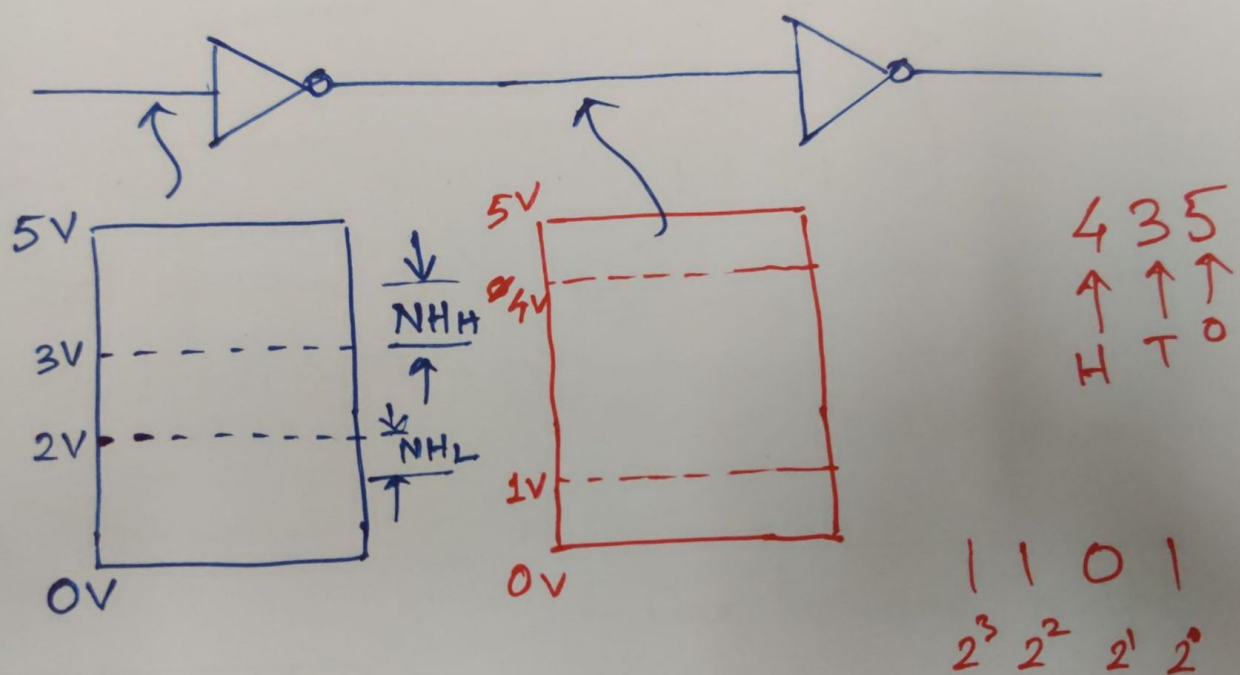
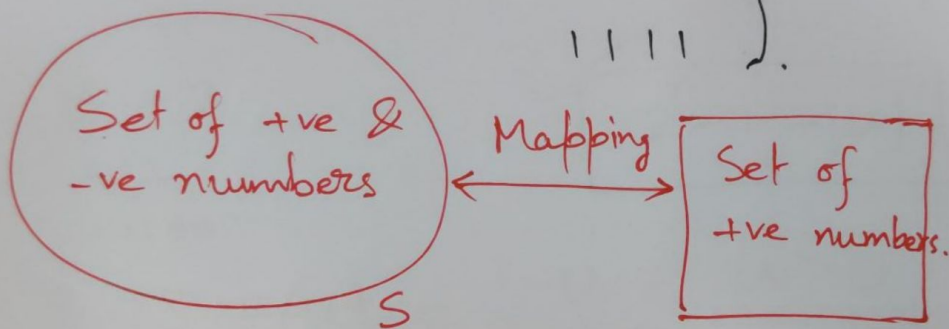




Negative Integers.

4 bit . 0000 }
 : }
 1111 }



sign bit	u
----------	---

$$\text{Sign bit} = \begin{cases} 1, & u < 0 \\ 0, & u > 0 \end{cases}$$

1. One-to-one mapping
2. All entries should be mapped.
3. Easy to perform addition/subtraction etc..

$$\begin{aligned} 5 &\rightarrow 0101 \\ -5 &\rightarrow 1101 \end{aligned}$$

1's complement

$$\begin{array}{rcl} 3 & \rightarrow & 0011 \\ -3 & \rightarrow & 1100 \end{array}$$

$$\begin{array}{cc} x & \bar{x} = 1 - x \\ 1 & 0 \\ 0 & 1 \end{array}$$

$$\begin{array}{l} 15 - 3 \\ = 12 \\ = 1100 \end{array}$$

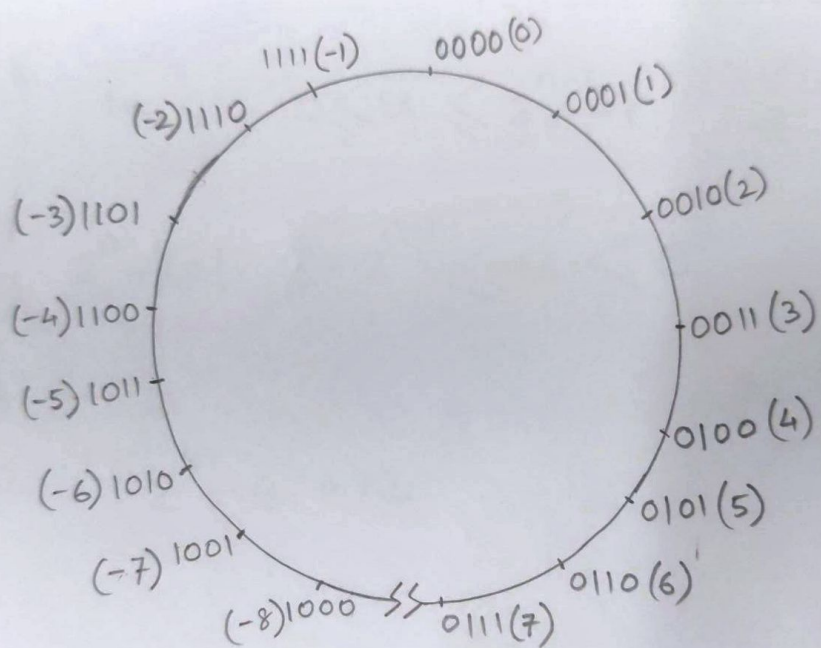
$$\boxed{2^n - 1 - |u|}$$

$$F(u) = \begin{cases} u, & u > 0 \end{cases}$$

-7 to 8

$$-3 + 7 = 4 \rightarrow 0100$$

$$8 + 7 = 15 \rightarrow 1111$$



2's complement Notation

$$F(u) = \begin{cases} u, & 0 \leq u \leq 2^{n-1} - 1 \\ 2^n - |u|, & -2^{n-1} \leq u < 0 \end{cases}$$

$0 \leq u \leq 7$
 $-8 \leq u < 0$

$$n = 4$$

$$4 \leftarrow 0100$$

$$-4 \leftarrow 2^4 - 4 = 12 = 1100$$

Properties.

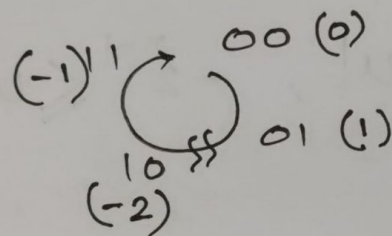
- * -2^{n-1} to $2^{n-1} - 1$
- * Unique representation of 0 (0000)
- * MSB is equal to sign bit.

$$\begin{array}{ccc} 1 & 1 & 0 & 1 \\ 2^3 & 2^2 & 2^1 & 2^0 \end{array}$$

Number Circle

2 bit

$$5 + (-2)$$



$$M + N$$

$$5 \quad (-2)$$

$$2^n - N$$

$$2^4 - 2 = 14$$

Computing 2's complement

$$* \quad 2^n - u$$

$$= \underbrace{2^n - 1 - u} + 1$$

↑
1's complement.

$$\boxed{2 \times (-3)} =$$

↓
0010

↓
1101

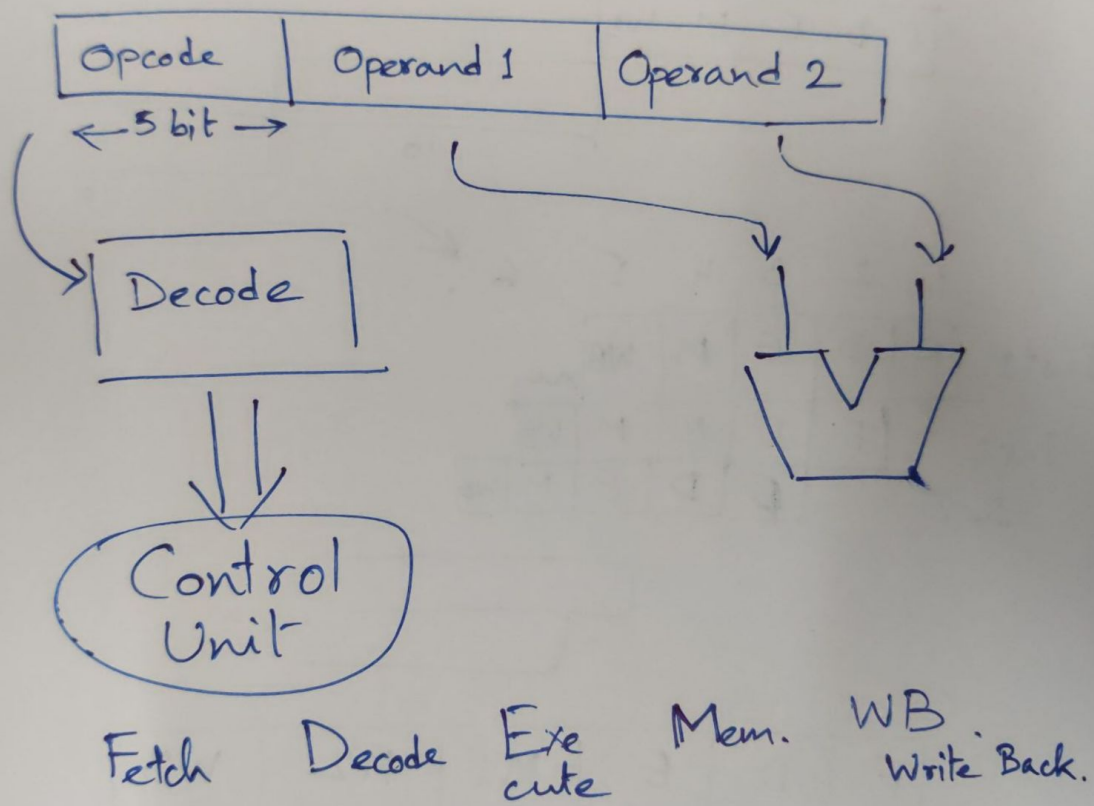
$$= 26 \bmod 2^4 = 10 \\ = 1010$$

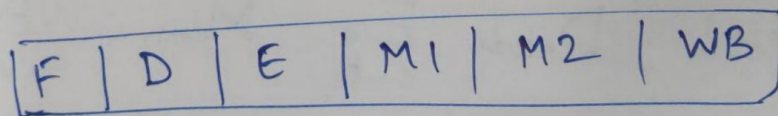
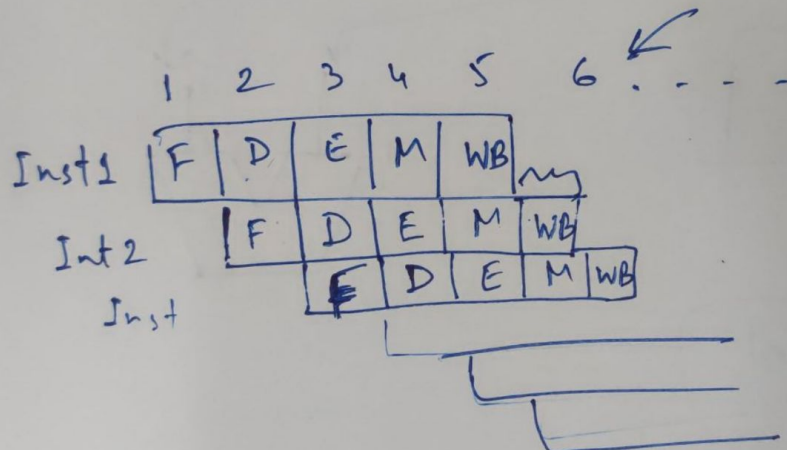
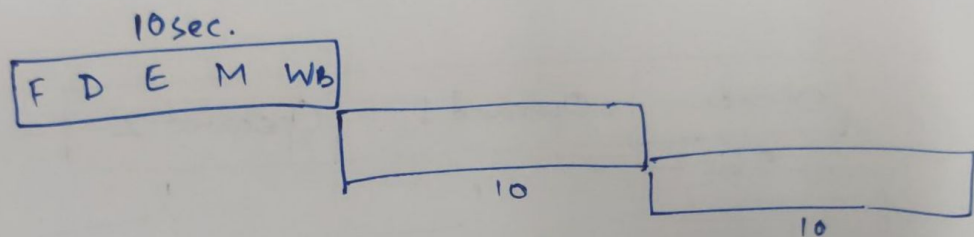
$$a \equiv b$$

$$\boxed{a = b \bmod 2^n}$$

$$a = 3$$

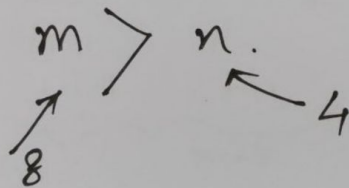
$$b = 19 \bmod 16 \\ = 3$$





Sign Extension

Convert a n bit number to a ~~mi~~
 m -bit number (2's complement)



+ve
Add $(m-n)$ 0's in the MSB positions.

0100 $\rightarrow$ 00000100

-ve:

$$F(u) = 2^n - |u|$$

$$F'(u) = 2^m - |u|$$

$$2^m - u - (2^n - u)$$

$$= 2^m - 2^n$$

$$F_4(-4) = 2^4 - 4$$

$$= 1100$$

$$= 2^n + 2^{n+1} + 2^{n+2} \dots + 2^{m-1}$$

$$= \underbrace{1111111}_{m-n} \underbrace{0000\dots 0}_n$$

$$F_8(-4) = 11110000 + 1100$$

$$= 1111\underbrace{1100}$$

To convert a -ve number

Add $(m-n)$ 1's in the MSB positions.

* In both cases extend the sign bit by $m-n$ positions.

Overflow theorem:

$u + v$

$u \cdot v < 0$, there will never be an overflow.

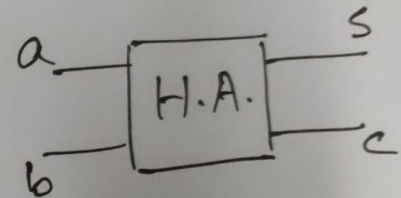
$u \cdot v > 0$, an overflow is possible.

u & v are of same sign.

→ the sign of the result is different from u & v then there is overflow.

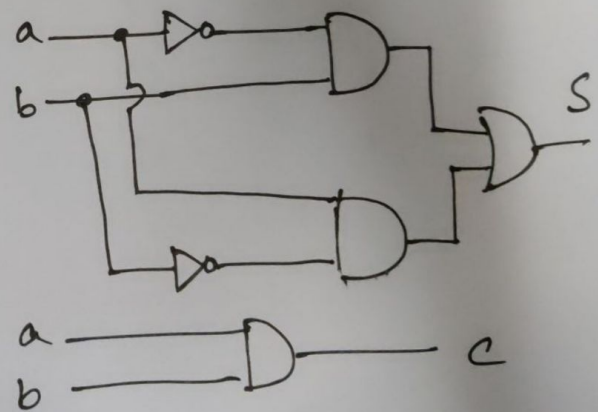
Truth Table of Half adder

a	b	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



$$S = \bar{a}b + a\bar{b}$$

$$C = a.b$$



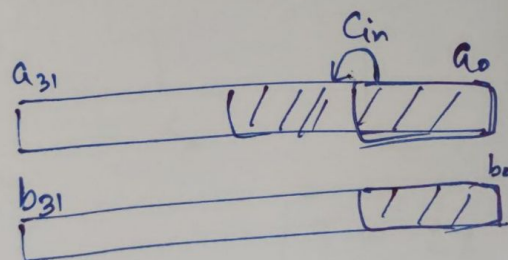
FA truth table

<u>a</u>	<u>b</u>	<u>cin</u>	<u>S</u>	<u>cout</u>
0	0	0	0	0
0	1	0	1	0
1	0	0	1	0
1	1	0	0	1
0	0	1	1	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	1

$$S = a \oplus b \oplus C_{in}$$

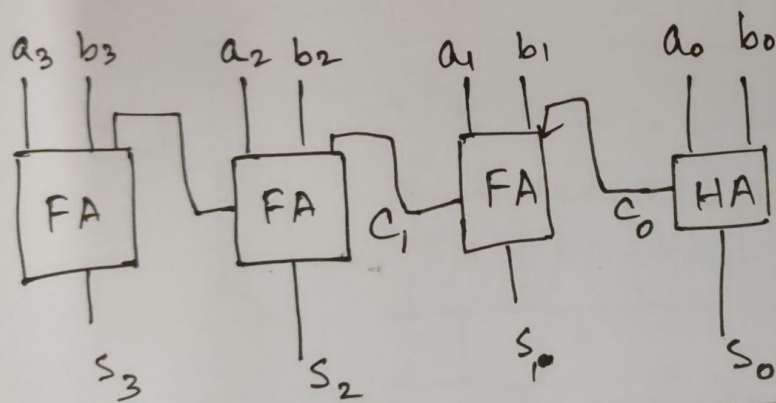
$$= a \cdot \bar{b} \cdot \bar{C}_{in} + \bar{a} \cdot b \cdot \bar{C}_{in} + \bar{a} \bar{b} C_{in} + a \cdot b C_{in}$$

$$C_{out} = a \cdot b + a \cdot C_{in} + b \cdot C_{in}$$

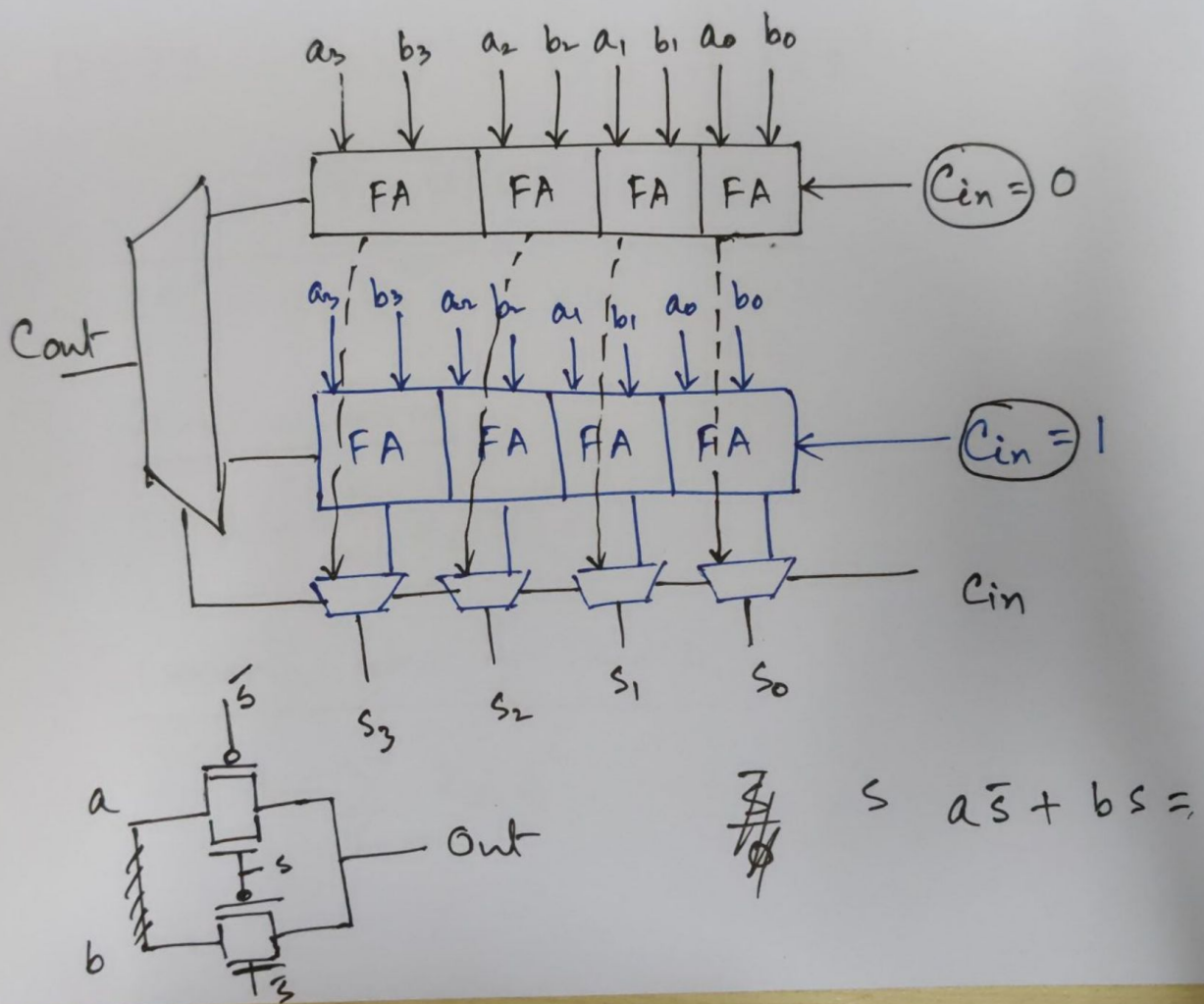


A : $a_3 a_2 a_1 a_0$

B : $b_3 b_2 b_1 b_0$



$$T = t_h + (n-1)t_f$$



$$0.375 = 0 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3}$$

$$\Rightarrow 0.011$$

$$3.29 = 3 \times 10^0 + 2 \times 10^{-1} + 9 \times 10^{-2}$$

Generic form

$$A = \sum_{i=-n}^n x_i 10^i$$

Generic form for base 2

$$A = \sum_{i=-n}^n x_i 2^i$$